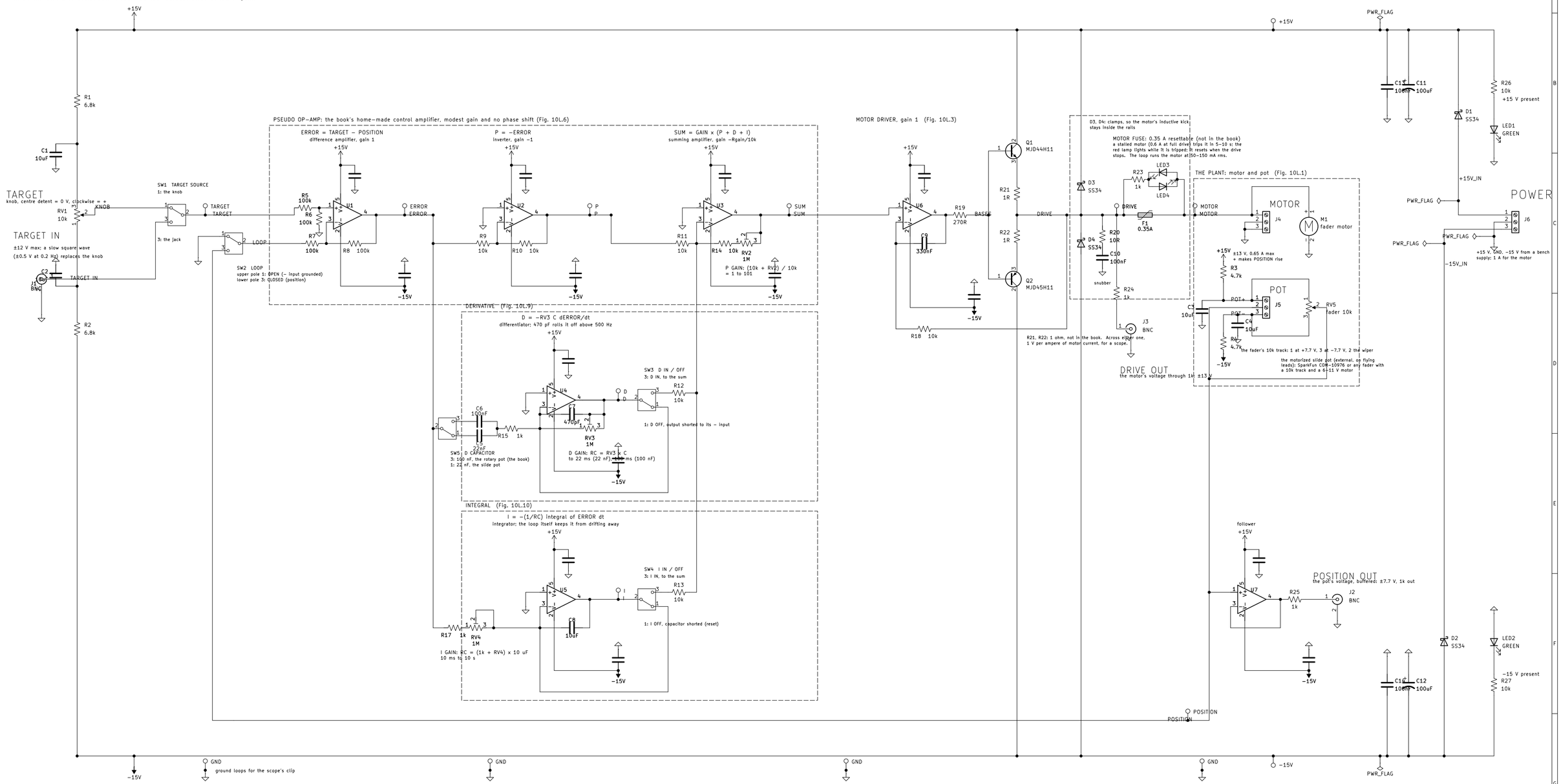
PID MOTOR CONTROL LOOP

Lab 10L of Learning the Art of Electronics (Hayes & Horowitz, 2016): a potentiometer sets a target position, a DC motor drives a second potentiometer to match it, and the controller is the book's home-made 'pseudo op-amp' of modest, adjustable gain, with the derivative and the integral of the error summed in. The board is the book's Fig. 10L.10 with every amplifier its own LM321 (the single LM358), the book's MJE3055/2955 as DPAK MJD44H11/45H11, and its resistor substitution boxes as trimmers.

Why a home-made op-amp: the motor integrates (voltage in, position out: −90 degrees from DC), so an ordinary op-amp's own −90 degrees would make −180 at a few hertz. The pseudo op-amp has gain 1 to 101 and no phase shift of its own. Simulated as drawn (sim/pid_sim.py, the fader modelled from its data sheet): P only rings at 12 Hz when the gain nears 100; D equal to P there wants RC = 1 / 2 pi f = 13 ms (RV3 near 611k with 22 nF), and then gain 100 settles in 90 ms with 1 % overshoot. I removes the residual error at low P gain.



TARGET
Knob or jack, on the TARGET SOURCE switch. The knob's 6.8k ends keep it to ±6.4 V, inside what the motor pot can reach (±7.7 V): a target the loop can always meet. A ±0.5 V square wave at 0.2 Hz from a generator on the jack gives repeatable steps. TARGET IN ±12 V max.

LOOP (SW2)
CLOSED: the pot's voltage into the difference amplifier. OPEN: its − input grounded, so the motor runs to an end stop as the knob leaves zero (the book's first test). Watch the fuse lamp: a stall trips it.

PSEUDO OP-AMP (U1, U2, U3)
U1 takes TARGET − POSITION with four 100k. U2 inverts it so P, D and I all arrive inverted at U3, which sums them through 10k each and multiplies by RV2's (10k + RV2)/10k = 1 to 101: the P GAIN. All three op-amps work at low closed-loop gain, so the chain has no phase shift where the loop's crossover is (a few hertz). At P gain 1.5 the loop is stable and slow; near 20 it overshoots; near 100 (P only) it oscillates; the period of that 'natural oscillation' sets D below.

Op-amps are drawn + on top: that is how the SOT-23-5 sits.

DERIVATIVE (U4, SW3, SW5)
D = −RV3 C dERROR/dt, rolled off above 500 Hz by the 470 pF. Set RV3 C to 1/6 of the natural period: for the slide pot (T of 40–90 ms) that is 6–14 ms, 300k–640k with the 22 nF; for the book's rotary motor-pot (T 0.6 s) use 100 nF and 1M. SW5 picks the capacitor. SW3 puts D into the sum or shorts it out.

INTEGRAL (U5, SW4)
I = −(1/RC) integral ERROR dt, RC = (1k + RV4) x 10 uF. OFF shorts the capacitor, so I starts from zero. I drives the residual error to zero; too much I rings.

MOTOR DRIVER (U6, Q1, Q2)
The book's Fig. 10L.3: the op-amp drives the push-pull through 270 ohm and the 10k takes the feedback from the motor side, so crossover distortion is inside the loop: the 0.33 uF feeds the op-amp's own output straight back so the phase-shifting transistors are bypassed at high frequency (split feedback, Lab 9L). 10 ohm + 0.1 uF snub the brush hash. Output ±13 V: the motor is rated 10 V, 6–11 V running, and sees full drive only in a step.

FUSE: not in the book. The 0.35 A PTC opens a stall (0.6 A) in seconds and a short in well under one, then resets itself when it cools. 1 ohm per emitter reads the motor current: 1 V per A across R21 or R22.

THE PLANT
Any motorized fader with a 10k track and a 6–11 V motor. Wire POT 1 to the track end that must read +7.7 V, 3 to the other, 2 to the wiper; MOTOR + and − so that a positive drive moves the wiper toward POT 1. If the loop runs away, swap the two motor wires. The second track and the touch pin of the SparkFun fader stay unused. POWER: ±15 V, 1 A (the motor's stall current). Series Schottkys against a reversed supply; 100 uF a rail for the motor's pulses; a green LED per rail. Quiescent about 10 mA. 100 nF at every op-amp pin.

Open source, CERN-OHL-P-2.0 -- github.com/gallicchio/motor-pid
P from 1 to 101, D up to 10 ms (slide pot) or 100 ms (rotary), I up to 10 s
LM321 (the single LM358) in SOT-23-5, one per triangle; MJD44H11 / MJD45H11 push-pull
Circuit: Hayes & Horowitz, Learning the Art of Electronics (2016), Lab 10L | Board: Jason Gallicchio

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